

Weeks 9–10: Evaluating Transparency, Explainability, and Accountability in Generative AI Systems

Introduction

Weeks 9 and 10 emphasize the increasing need for explainability and transparency in generative AI. The demands for accountability and clarity increase as these models play bigger roles in writing, analysis, decision support, and automation. This section examines explainability for systems like ChatGPT, how AI companies communicate model behavior, and the importance of accountability frameworks for safe deployment.

1. Understanding Transparency in AI Models

Transparency is the openness with which an organization explains how its model operates, what data it uses, and how decisions are made. Complete transparency in LLMs is challenging due to:

- Sources of training data are numerous and frequently proprietary.
- The internal workings of the model are complicated and difficult to understand.
- Businesses must strike a balance between security, safety, and intellectual property issues and openness.

As a result, transparency frequently manifests as:

- System cards that describe the risks and behavior of the model
- Model documentation outlining the capabilities and constraints
- Policy disclosures outlining operational limitations and safety regulations
- High-level summaries of the categories of training data

Transparency helps users understand how to responsibly interact with the model, even though it cannot reveal every detail.

2. Explainability in Generative AI

The goal of explainability is to assist users in comprehending why a model generated a particular result. LLMs' "reasoning" is a combination of pattern recognition and probabilistic generation, in contrast to traditional machine-learning models, which have straightforward decision boundaries.

Among the explainability techniques used today are:

- **Analysis of token-level probability**
displaying the terms or concepts that the model most likely predicted.
- **Summaries of chains of thought**
simplifying reasoning without revealing unprocessed internal workings.
- **Visualization of attention patterns**
highlighting the aspects of the input that affected the model's output.
- **Behavioral justifications**
describing the known failure modes, boundaries, and tendencies of the model.

Giving users clarity without overburdening them with technical details is the goal of explainability.

3. Accountability in Deployment

Accountability establishes who oversees model behavior: developers, deployers, or end users. This has emerged as a major topic of discussion in AI governance.

Important accountability factors consist of:

1. Preventing Misuse

Businesses establish usage guidelines to stop detrimental applications like harassment, impersonation, and false information.

2. Monitoring and Reporting Errors

Organizations keep organized systems in place to monitor detrimental outputs and make iterative improvements.

3. Adherence to Regulations

AI providers must abide by laws like:

- GDPR (EU)
- The EU AI Act
- The California Consumer Privacy Act (CCPA)
- Framework for NIST AI Risk Management

4. Explicit User Contracts

Platforms convey expected usage conditions, liability limits, and user responsibilities.

Accountability ensures that AI models are implemented safely and consistently by combining operational, ethical, and legal requirements.

Table: Transparency, Explainability, and Accountability Overview

Area	What It Means	Examples in Practice
Transparency	Provide information about how the model works, its limitations, and data scope	System cards, data summaries, safety disclosures
Explainability	Help users understand why a model produced a given output	Token probabilities, reason summaries, behavior notes
Accountability	Clear assignment of responsibility for errors, misuse, and oversight	Usage policies, compliance requirements, audit logs

4. Social Impact Factors

Assessing accountability and transparency has an impact on public trust in addition to being technical. Insufficient clarity may result in:

- Inaccurate interpretation of model results
- Overuse of AI in delicate decision-making
- Confusion about accountability when mistakes are made
- Strong accountability procedures, responsible disclosures, and clear communication all aid in lowering these risks.

5. Conclusion

The understanding, trust, and application of generative AI systems are shaped by transparency, explainability, and accountability. Large language models' internal workings are complicated and difficult to fully understand, but responsible governance practices, meaningful explanations, and clear documentation can help close the gap between technical constraints and user expectations. During these two weeks, the emphasis will be on identifying how businesses convey model behavior, how explainability tools assist users in understanding outputs, and how accountability frameworks specify roles for developers, deployers, and users. When combined, these factors facilitate the safer and more dependable use of generative AI.

Timeline

- Week 1-2 (10/20/2025 - 11/04/2025)

Deliverables (Weeks 9–10)

- A written summary analyzing transparency, explainability, and accountability in LLMs.
- A comparison table outlining their roles and practical implications

Reference

- <https://openai.com/research/>
- <https://www.unesco.org/en/artificial-intelligence/recommendation-ethics>
- <https://www.nist.gov/itl/ai-risk-management-framework>
- <https://www.nature.com/natmachintell/>

